CSCI 272 Formal Languages **Spring 2025 - Homework Exercise 3**

This homework assignment is worth 22 points total.

Solve the following problems to the best of your ability, writing your solutions by hand on white paper. Don't write too small, and use clear handwriting. You may use a pencil or pen, but not one that uses red ink.

You will need to submit your solutions both physically and digitally. The digital copy should be submitted to Moodle by Tuesday, March 25 by 23:59. The hard copy will be due on Wednesday, March 26 just before the start of your lecture section. The physical copy must be identical to what was submitted digitally.

Scanning Guidelines:

- Make a good quality scan. Phone pictures (CamScanner) are acceptable, as long as they are in focus and readable.
- Before submitting, verify that you can read your own scans.
- Scan the pages in the correct orientation (not rotated over 90 degrees.)
- You may submit your scans in a single .pdf file with the solutions presented in the same order as the questions, or collect them into a single archive file (.zip, .gzip, or .rar) with your files numbered in such a way that they appear in the right order.

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1. (16 points total) For each of the following, create a Turing Machine that meets the given description. Solutions must be given as an automata diagram using the definition of TM given in the class. However, you are encouraged to check your answers by writing the textual definition of the TM in an .ntm file and running them in Hans' simulator.

a) (3 pts.) A Turing Machine that decides the language of palindromes over the alphabet {a, b, c}. Don't forget that palindromes can have odd or even length, and that the empty string is also a palindrome.

b) (3 pts.) A Turing Machine that decides the language of strings over the alphabet {a, b} which has the same number of a's and b's. The a's and b's can be in any order. Note that the language includes the empty string.

c) (4 pts.) A Turing Machine that decides the language:

$$L = \left\{a^i b^j \,\middle|\, j = 2^i \text{ and } j \geq 1, i \geq 0\right\}$$

d) (2 pts.) A Transducer Turing Machine that takes strings of binary digits (the characters 0 and 1) as input, and increments the represented number by one.

A **Transducer Turing Machine (TTM)** is a variation of deterministic Turing Machine that is used to define functions on strings. The starting input string on the tape of a TTM is the input to the function, and then the TTM makes modifications to the string during the computation. When the TTM enters an accept state, it immediately stops, and the resulting string that is on the tape defines the resulting output of the function.

e) (3 pts) A Transducer Turing Machine that takes strings over the alphabet {a, b, c} as input, and then outputs the reversed string.

2. (7 points total) A language L is called **decidable** if there exists a Turing Machine that recognizes the language L, and is guaranteed to terminate, i.e., for any input it reaches an accepting state or gets "stuck" resulting in rejecting the input word – it can never get into an infinite loop.

Using this definition, justify the following true statements by describing how you would construct relevant Turing Machines from those Turing Machines that decide the given languages A and B.

a) (2 pts.) If you have languages A and B that are both decidable, then $A \cup B$ is also decidable.

b) (2 pts.) If you have languages A and B that are both decidable, then the language $A \cap B$ is also decidable.

c) (3 pts.) If you have languages A and B that are both decidable, then the language $A \circ B$ is also decidable. Recall that $A \circ B = \{ w_1 w_2 \mid w_1 \in A, w_2 \in B \}$.